Anqi Tan

Ph.D. Student, Bioengineering · Seeking 2027 Internship in AI & Data for Health

Urbana, IL | anqitan2@illinois.edu | anqitan.github.io | LinkedIn | Google Scholar

Builds data pipelines, VLM agents, and LLM tools around real biomedical devices and human-sample data. Available May–Aug 2027 (on-site or remote) and part-time remote during the school year.

EDUCATION

Ph.D. in Bioengineering, University of Illinois Urbana-Champaign Aug 2022 – Expected Dec 2027

Advisor: Prof. Brian T. Cunningham · Nanosensors Group

B.S. in Mechanical Engineering (dual degree), UIUC & Zhejiang University Aug 2018 – Jun 2022

EXPERIENCE

Graduate Research Assistant, Nanosensors Group, UIUC Sep 2022 – Present

- Built a reader image-analysis pipeline (**test/control-line detection, background correction, 4PL calibration**) that turns strip images into cortisol concentrations in an Android app (Kotlin).
- Developing a **quantification-aware active-learning** method that reduces expert labeling for particle counting across biosensing imaging platforms (first-author manuscript in preparation).
- Lead the Live Hormone project: a salivary cortisol lateral flow test (**LOD ≤0.12 ng/mL**) with a phone-controlled reader, tested with **153 adults** (IRB-approved); **oral presentation, APHA 2026**.
- Lead reader development for a **$4.87M DoD (PRMRP)** point-of-care sepsis/ARDS project with Stanford, WashU, and UCSF; reader now in use at WashU on **patient urine samples**.

Research Assistant Intern, Institute of Brain Science, BrainCo (Hangzhou, China) May 2022 – Aug 2022

- Conducted literature and competitor reviews, recruited volunteers, and collected data for a sleep-aid stimulator and an EEG fatigue monitor.

Undergraduate Research Leader, ZJU-UIUC Institute 2019 – 2021

- Led a national-level undergraduate research project designing a re-localizable remote-center-of-motion mechanism for minimally invasive surgery (**first-author paper, IEEE ICARM 2021**).

SELECTED PROJECTS

RETICLE: Pathology VLM Agent (co-first author) 2026 – Present

- Built the expert models, retrieval over 98K case reports and 592K images, agent framework, and evaluation; lifted metastasis-sizing and mitosis-counting accuracy from near chance (25–31%) to **66–72%**.

CRISPR Mechanism Explainer (Ashby Prize Hackathon, NCSA) — 1st Place Oct 2025

- Found label bias in the public benchmark, rebuilt a balanced validation set, and fine-tuned an LLM (Qwen3) to classify CRISPR screen outcomes and explain the mechanism; shipped a Gradio demo.

PUBLICATIONS

First author: 3 manuscripts in preparation (LumaPF cortisol reader; active learning for particle counting; plasmonic LFA labels) · IEEE ICARM 2021. **Co-first author:** RETICLE (in preparation). **Co-author:** 4 journal papers (*Chemical Reviews*, *Lab on a Chip*, *ACS Nano Medicine*, *Biosensors*) and 1 Elsevier book chapter. **Talks:** APHA 2026 (oral), LOAC 2026.

HONORS & TEACHING

1st Place, Ashby Prize Computational Science Hackathon, NCSA (2025) · National Silver Medal, China "Internet+" Innovation & Entrepreneurship Competition (2021) · TA, BIOE/ECE 415 Biomedical Instrumentation Lab (2025) and ME 170 Computer-Aided Design (3 semesters)

SKILLS

AI & Data: Python, PyTorch, vision-language models, RAG and LLM agents, LLM fine-tuning (Qwen3), active learning, A100 GPU cluster, OpenAI API, Gradio, image processing, MATLAB

Biosensing: lateral flow assays (competitive & sandwich), plasmonic-fluor labels, immunoassay optimization, ELISA, saliva/plasma sample prep, fluorescence imaging, IRB human-subject studies

Hardware: Raspberry Pi, ESP32, Arduino, LabVIEW, IMU sensors, fluorescence optics

Languages: English (fluent), Chinese (native)